Section I: Aims and Background

Most armed conflicts leave behind landmines—or unexploded ordnance (UXO)—which pose a tremendous hazard. These devices are deliberately designed to remain hidden from human eye or photography of any kind, either buried beneath the soil or concealed in dense vegetation. Governments worldwide invest heavily in both technological and manual demining approaches, since UXO threatens lives, communities, and economic activity.

Although demining still relies primarily on humans and manual labour, recent efforts have explored ground-based autonomous systems. In practice, however, rugged terrain and dense vegetation often limit their effectiveness.[1] To overcome these challenges, drone-based thermal imaging has emerged as a promising alternative. A key difficulty remains the reliable extraction of relevant information from complex imagery; to illustrate this, Figure 1 presents two samples of surface-level landmine detections captured at an altitude of 0.8 m.

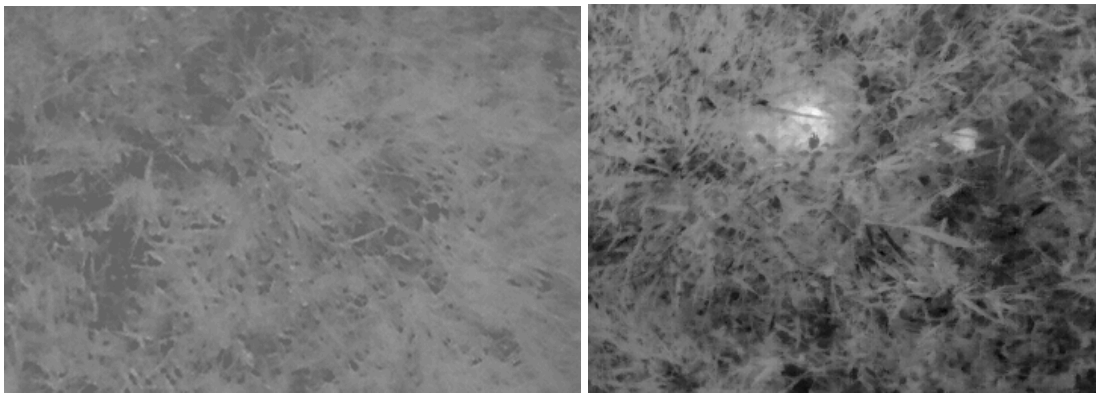


Figure 1. Two surface-level landmine images captured at an altitude of 0.8 m, with central temperatures of 30 °C and 25 °C, respectively.

Our objective is to develop a convolutional neural network (CNN) model, fine-tuned for landmine detection—optimizing the F1 score with a particular emphasis on minimizing false negatives.

For model training, we use a dataset of thermal images acquired with a Zenmuse XT infrared camera (7–13 μm) mounted on a DJI Matrice 100 drone, provided by a research team in Cali, Valle del Cauca, Colombia. In this dataset, landmines are buried at depths of 0 cm, 1 cm, 5 cm, and 10 cm across nine 10 m × 10 m test zones, along with a smaller collection of mine-free images.[2]

[1] Davor Antonic, “Demining Robots – Requirements and Constraints,” in Proceedings, June 2001.
[2] Alejandro Tenorio Tamayo et al., *Test Images of Buried Landmines*, version 4, Mendeley Data, 2023, https://doi.org/10.17632/732ngnf4r3.4.

Section II: Contributions of Team Members

We gratefully acknowledge Hermes Alejandro Tenorio-Tamayo, Juan Camilo Forero-Ramírez, Bryan García, Humberto Loaiza-Correa, Andrés David Restrepo-Girón, Sandra Esperanza Nope-Rodríguez, Asfur Barandica-López, and José Tomás Buitrago-Molina for providing the thermal image dataset used in this research. All data preparation, preprocessing, model training, and writing were carried out by Igor Klymenko.

Section III: Methodology

Overview

Our objective is to employ the You Only Look Once (YOLO) algorithm—a Single Shot Detector (SSD) convolutional neural network that processes each image once and outputs a probability for the “landmine” versus “no landmine” class. Our implementation incorporates C3K2 and C2PSA blocks, as well as Spatial Pyramid Pooling Fast (SPPF); these components are detailed in the Model Description section. Figure 2 illustrates the general YOLO architecture.[3]

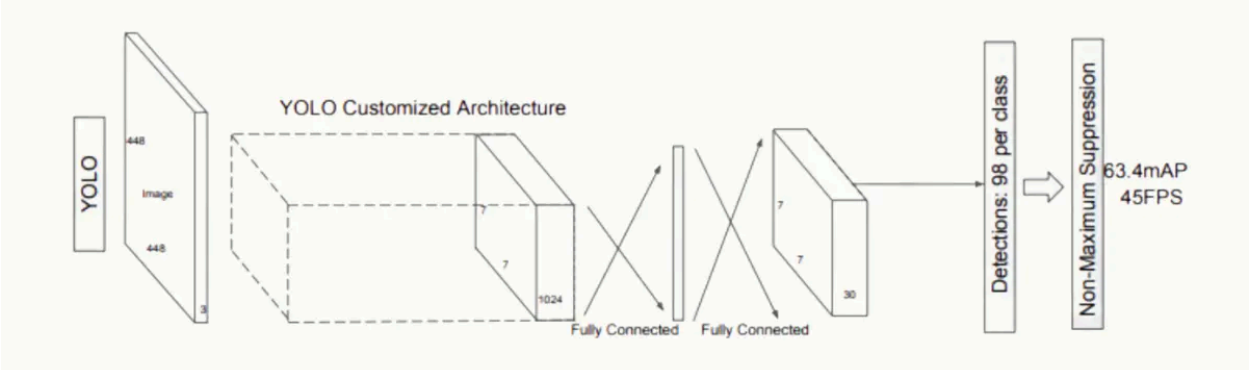


Figure 2. General structure of the YOLO model

In this chapter, we will first discuss data acquisition, preparation, and preprocessing, including the dataset’s distribution and augmentation we have done. Next, we will cover more detailed YOLOv11 model architecture, our data-splitting method, the fine-tuning process, hardware specifications, chosen hyperparameters, loss function, and evaluation metrics.

Data Acquisition[4]

The dataset was collected during flights over the Universidad del Valle facilities in Cali, Valle del Cauca, Colombia, using a Zenmuse XT infrared camera (7–13 μm) mounted on a DJI Matrice

[3] Rahima Khanam and Muhammad Hussain, “YOLOv11: An Overview of the Key Architectural Enhancements,” arXiv:2410.17725 [cs.CV] (October 2024), https://doi.org/10.48550/arXiv.2410.17725.
[4] Alejandro Tenorio Tamayo et al., *Test Images of Buried Landmines*, version 4, Mendeley Data, 2023, https://doi.org/10.17632/732ngnf4r3.4.

100 drone. Antipersonnel legbreaker–style mines, alongside mock objects of similar characteristics, were buried at depths of 0, 1, 5, and 10 cm across the test zones.

Raw thermal images were captured in three resolutions and formats:

- 720 × 480 px (JPEG)
- 336 × 256 px (JPEG)
- 336 × 256 px (TIFF)

For our experiments, we used only the first two formats. All images were acquired at altitudes between 1 and 10 m across nine distinct areas in 5 different flight missions.

Data Preparation

Originally, the dataset was split across five flight missions by date. Within each mission, data were organized by landmine burial depth, mine-free zones, and data type. For our experiment, we simply divided the images based on whether they contained a landmine, creating a diverse dataset over multiple central temperatures, burial depths, and flight altitudes. This decision is grounded in the model's SPPF mechanism, which applies multiple max-pooling operations with different kernel sizes to generate multi-scale contextual information. It therefore enables the model to pay attention to similar objects represented in different sizes and shapes—here, landmines captured at flight altitudes ranging from 1 to 10 meters. As mentioned earlier, for the experiment we focused only on JPEG images from the dataset.

Overall, the dataset comprised 870 images containing landmines (positive samples) and 108 images of "free zones" (negative samples) with an overall number of 978 samples.

Data Preprocessing

The first challenge we encountered was the extreme class imbalance: positive samples comprised 89.0% of the dataset, while negative samples accounted for only 11.0%, resulting in an imbalance ratio of approximately 8.1 and a Gini impurity of 0.20.

After the random split of the dataset into 70% for training and 30% for validation we applied the image augmentation on the negative set of samples in training using:

- Horizontal flip
- Random rotation by 90°, 180°, or 270°
- Random color jitter

Afterwards, the dataset included 870 positive samples and 333 negative samples, yielding an updated class distribution of 72.3% positive versus 27.7% negative, an imbalance ratio of approximately 2.61, and a Gini impurity of 0.40—indicating a substantially more balanced dataset. It resulted in 909 images for training and 294 images for validation.

Model Description

We chose a single-shot CNN with an attention mechanism to focus on small objects, following the YOLOv11 architecture. Below, we describe the used architecture in detail using three primary components:

- **Bottleneck:** Responsible for feature extraction. A sequence of convolutional blocks transforms input images into multi-scale feature maps.
- **Neck:** Aggregates and enhances feature representations across different layers.
- **Head:** Uses the refined features from the neck to predict object classes and bounding boxes.

Bottleneck

The bottleneck is composed of convolutional blocks, each of which may include a residual connection. A single convolutional block downsamples its input feature map (dimensions C×H×W) by applying:

1. A 2D convolutional layer
2. A 2D batch-normalization layer
3. A SiLU activation function

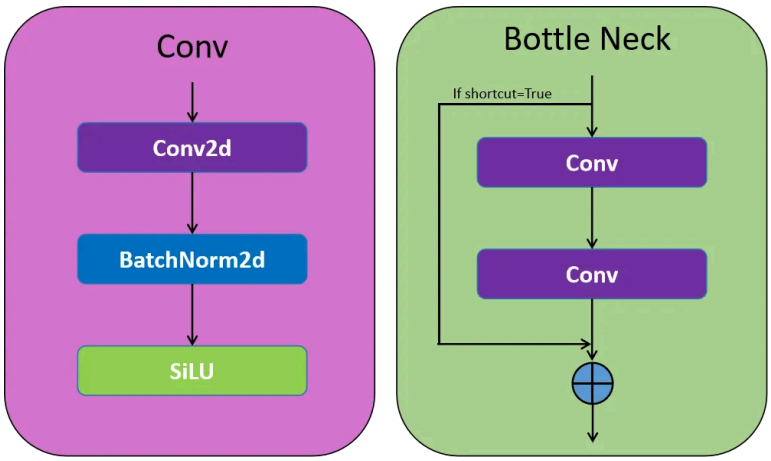


Figure 3. Convolutional and bottleneck blocks used in YOLO

YOLOv11 employs a C3K2 block—an enhanced version of the Cross Stage Partial (CSP) bottleneck. The C3K2 block uses multiple 3×3 convolutional layers as preprocessing steps to produce finer-grained feature maps. These smaller feature maps are concatenated and passed on to the next layer.

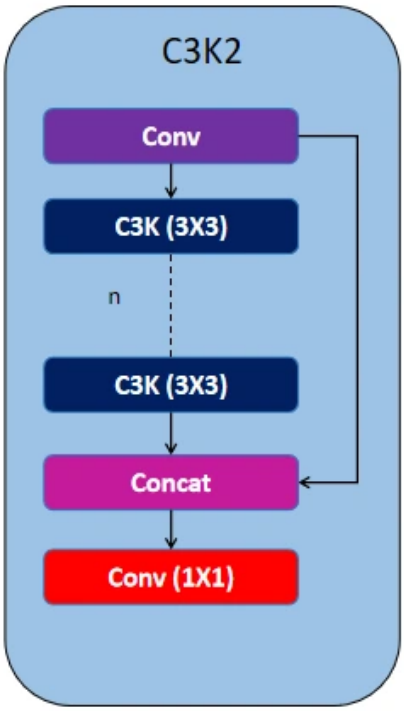


Figure 4. C3K2 block used in Bottleneck

Neck

Here, YOLOv11 incorporates the previously mentioned SPPF mechanism, which uses varied kernel sizes to extract contextual features via max-pooling. This component is essential for landmine detection because it enables the model to represent very small objects in the feature maps passed to the head for prediction.

The second, critical in given scenario, feature for detecting nearly indistinguishable thermal signatures of landmines is the attention mechanism, implemented in the Cross Stage Partial with Spatial Attention (C2PSA) block. The C2PSA block allows the model to emphasize spatial differences of small objects that might otherwise blend into the background. By applying spatial attention, the model can focus on localized regions of interest extracting features that later are passed into Head. See implementation image below:

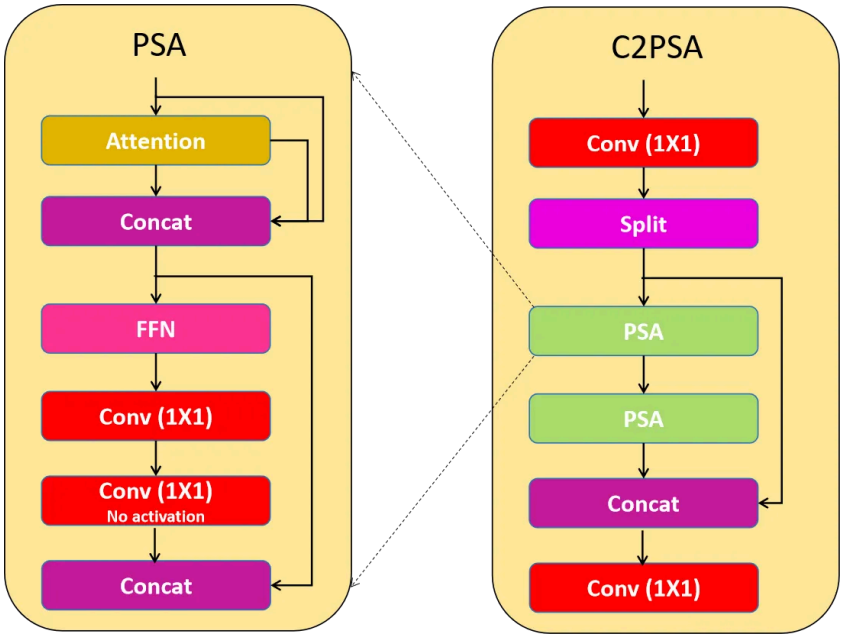


Figure 5. C2PSA block used in Neck

Head

Since our model needs to produce a single class prediction for the entire image, the head processes the neck's feature map through one convolutional layer followed by two fully connected layers, outputting a single probability that the image contains a landmine.

Loss Function and Optimizer

Since we only have two classes ("landmine" vs. "no landmine"), we use Binary Cross-Entropy (BCE) as the loss function. We use AdamW as an optimizer to avoid overfitting by the use of weight decay decoupled from the gradient update process.[5]

Section IV: Experiment Setup

All experiments were conducted using GPU T4 in Google Collab. We used a pre-trained YOLOv11n-CLS classification model as our base and fine-tuned it for landmine detection using the architecture and data described above.

We conducted three experiments, testing 0.3, 0.5, 0.8 dropout rates. In each run, we used a learning rate of 0.001429, a momentum of 0.9, a batch size of 32, resized inputs to 320 × 320 pixels, and trained for 20 epochs.

[5]Ilya Loshchilov and Frank Hutter, "Decoupled Weight Decay Regularization," arXiv:1711.05101 [cs.LG] (2019), https://arxiv.org/abs/1711.05101.

Section V: Evaluation and Results

As introduced earlier, the model incorporates the C2PSA attention mechanism, so the training images were specifically augmented to suit this architecture. To illustrate what data is used in training of the model, Figure 6 shows a single training batch.

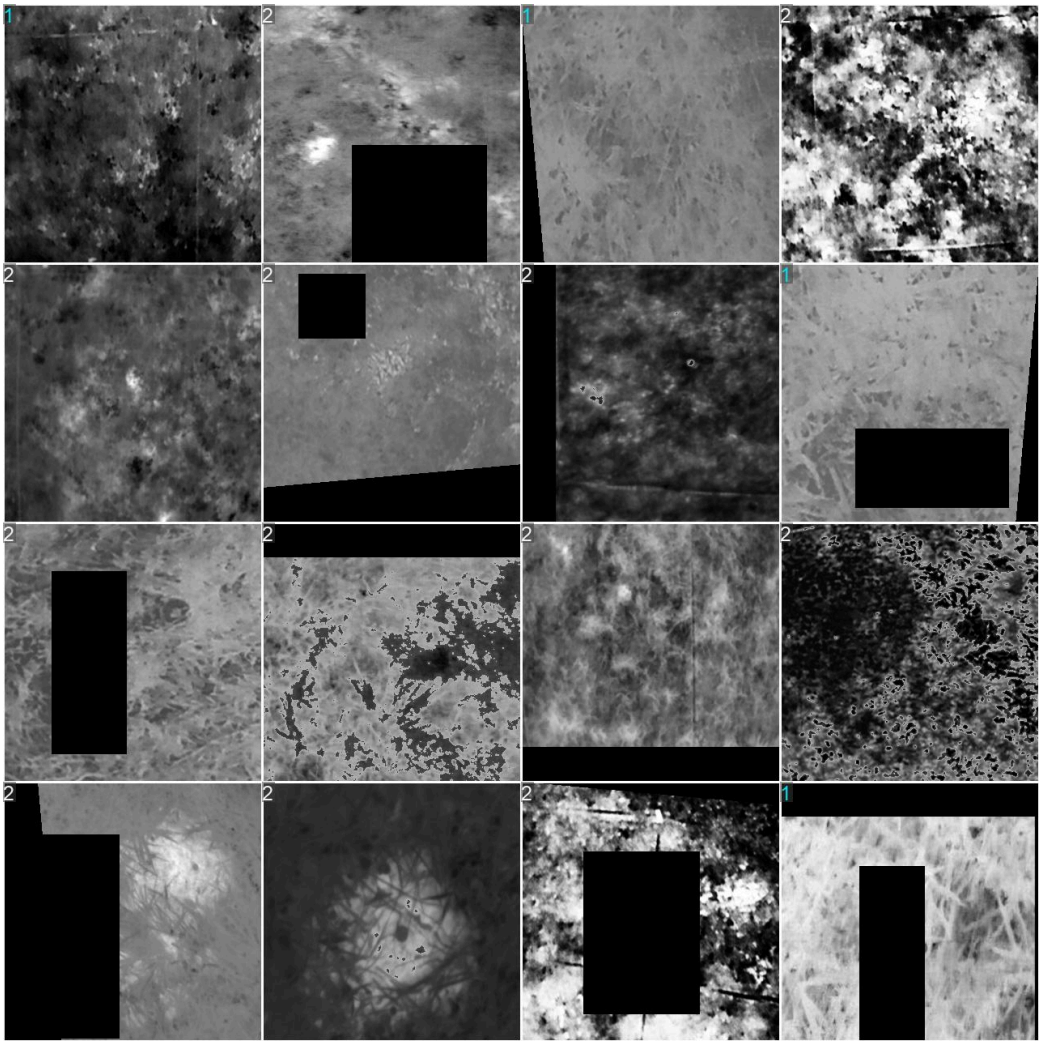


Figure 6. An example of a batch used for training.

After conducting our experiments, the best performance was achieved with a dropout rate of 0.3, yielding an accuracy of 98.29%. The next best results came from dropout rates of 0.8 and 0.5. Figure 7 plots the accuracy progression for each dropout setting.

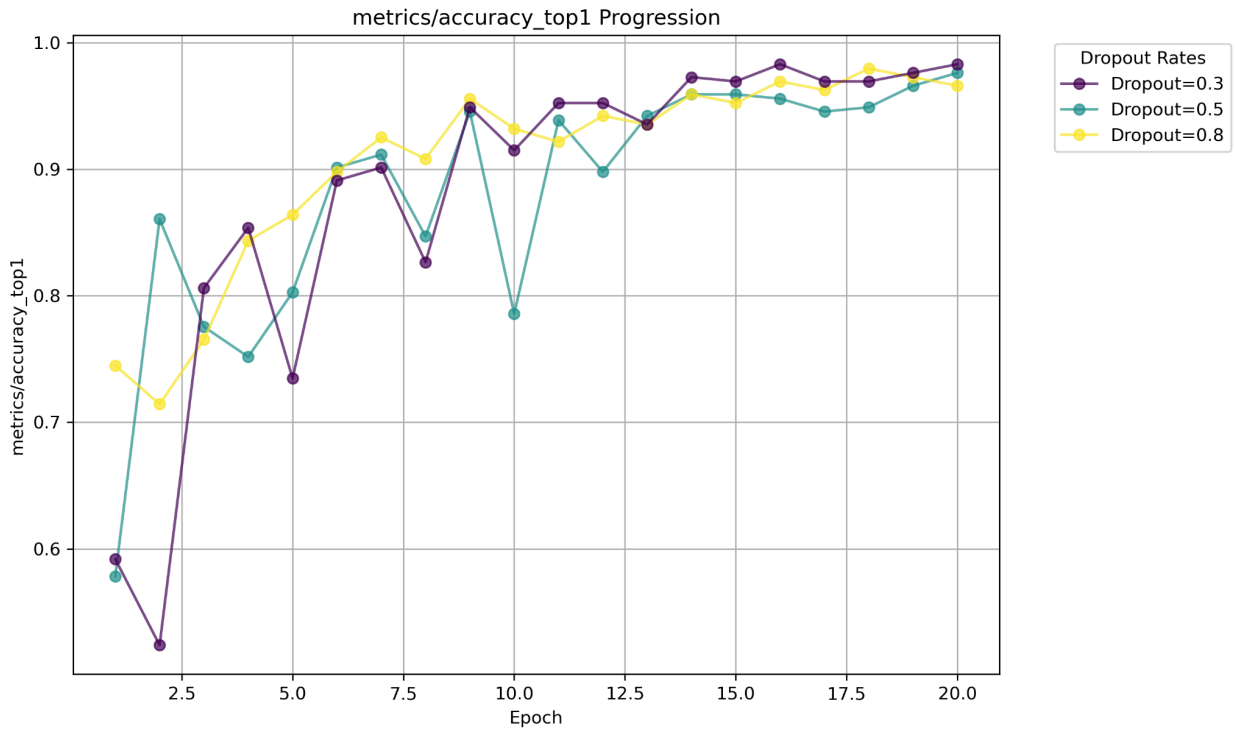


Figure 7. Accuracy progression for different dropout rates.

It is also important to note that, despite several anomalies during training in each run, the model did not overfit. This can be seen by comparing the loss curves for training and validation across experiments, shown in Figures 8 and 9 below.

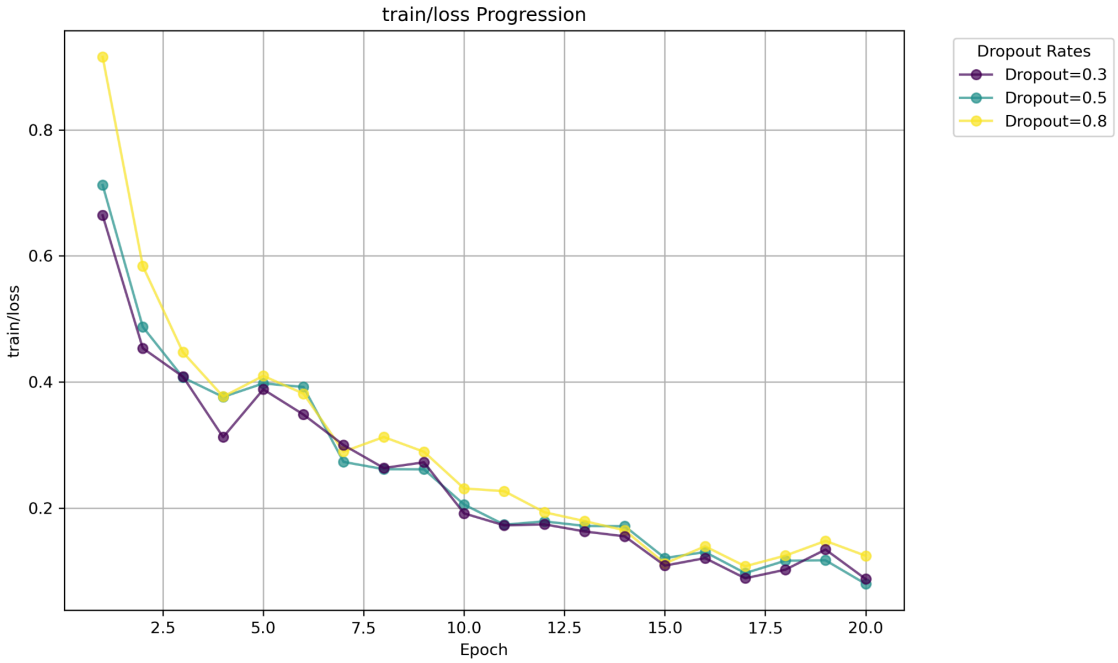


Figure 8. Loss during training progression for different dropout rates.

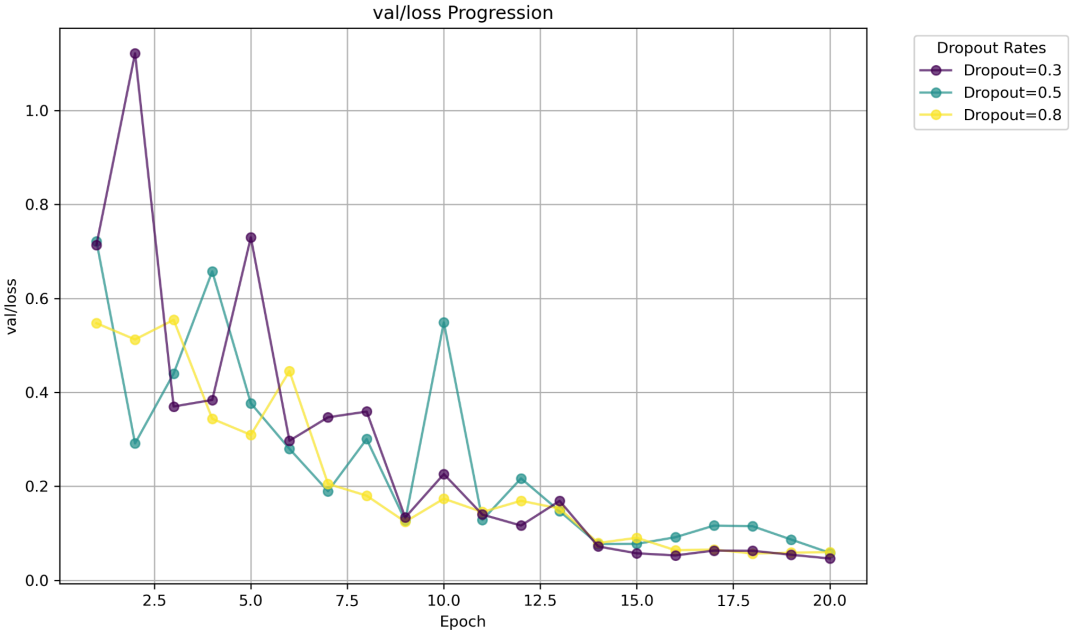


Figure 9. Loss during validation progression for different dropout rates.

In summary, the model performs well on this dataset, achieving 98% accuracy without signs of constant overfitting or labelling everything to positive class (as it did in the second epoch for dropout 0.3 in the Figure 9). Under this configuration, the normalized confusion matrix (Figure 10) shows 0.06 false negatives and only 0.01% false positives.

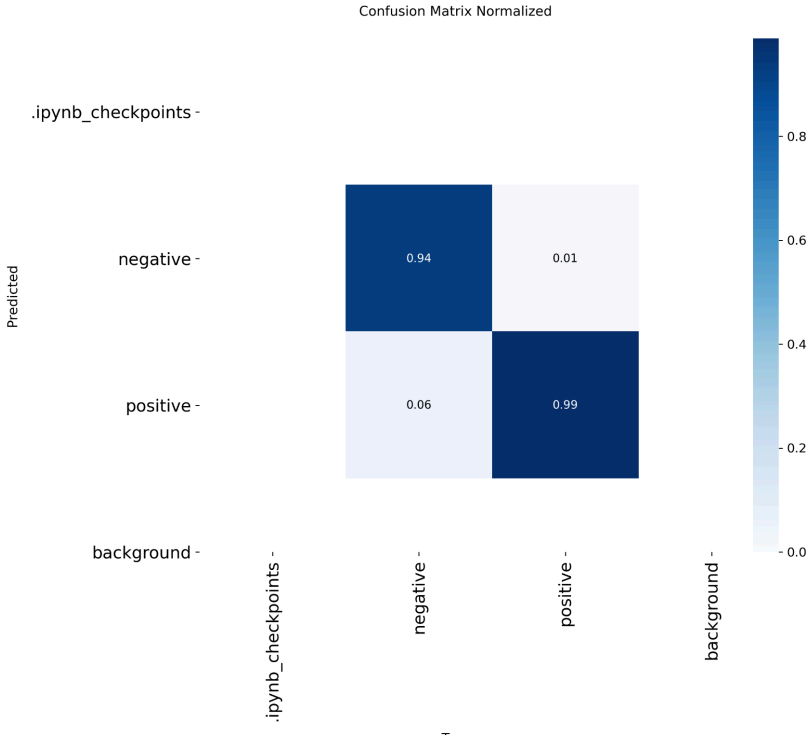


Figure 10. Normalized confusion matrix for a dropout rate of 0.3.

Finally, to give a clearer sense of the data the model processed, Figure 11 presents a selection of validation images that were correctly classified. In these examples, “positive” indicates the presence of a landmine.

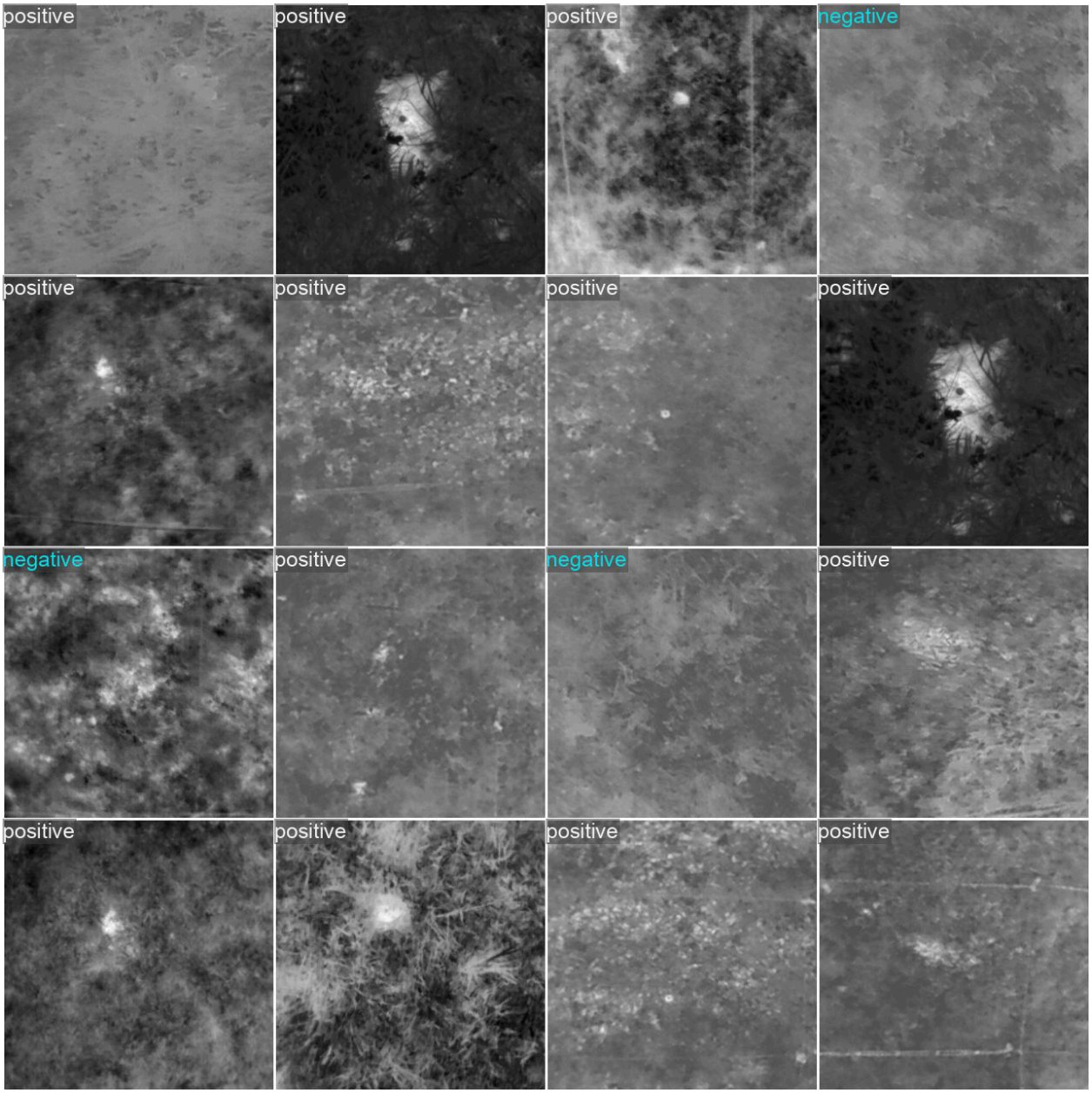


Figure 11. Examples of correctly predicted labels in the validation set.

Section VI: Conclusion

In this work, we trained an efficient model to classify thermal images (720 × 480 px and 336 × 256 px) captured by a drone flying at heights of 1–10 m, identifying potential legbreaker-style landmines buried up to 10 cm below the ground. One of the main challenges in this classification task is the diffusion of the landmine's thermal signature into its surroundings, which generates significant noise and makes such objects nearly unrecognizable to the human eye.

To tackle this, we employed the YOLOv11 architecture, fine-tuning YOLOv11n-CLS to classify images—taken from various heights over different landmines—as either containing a landmine or being a free zone. Mechanisms like SPPF and C2PSA blocks enabled the model to leverage multi-scale contextual representations and spatial attention, which are crucial for extracting the small, subtle features that distinguish landmine signatures from a homogeneous background.

Using the augmented data (augmenting only the free-zone images), we improved the class imbalance from an initial ratio of 8.1 to 2.61. Using a 70/30 random split (before augmentation), we ran three experiments with three dropout rates (0.3, 0.5, 0.8), using a batch size of 32, input size of 320 × 320, and 20 epochs per run. The best performance achieved was 93% accuracy, with 91.18% recall, 93.94% precision, and an F1 score of 92.54%.

Our results suggest that this approach holds promise for providing reliable classifications from drone-collected thermal images of landmines. However, because several factors were not specified by the data collection team—such as time of day, season, sunlight levels, and ambient temperature—we cannot claim that the model will generalize with equally high accuracy across all landmine-contaminated zones. More importantly, though, we can say that the model can distinguish landmine thermal signatures in extremely ambiguous scenarios using the described architecture and approach.

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4. Tenorio Tamayo, Alejandro, Juan Camilo Forero Ramirez, Bryan García, Humberto Loaiza Correa, Andrés David Restrepo Girón, Sandra Esperanza Nope, Asfur Barandica López, and José Tomás Buitrago. *Test Images of Buried Landmines*. Version 4. Mendeley Data, 2023. https://doi.org/10.17632/732ngnf4r3.4.